

EXP.NO: 4	System calls-fork(),exec(),getpid(),opendir(),readdir()
DATE:	

AIM:

To study and implement various Unix system calls to understand process creation, execution control, process identification, and directory operations.

ALGORITHM:

1. Start the program.
2. Include necessary header files (stdio.h, unistd.h, dirent.h, etc.
3. Process Creation using fork()
 - Call fork() to create a child process.
 - Check the return value of fork():
 - i. If 0 → Child process executes.
 - ii. If > 0 → Parent process executes.
 - iii. If < 0 → Process creation failed.
4. Process Identification using getpid()
 - Use getpid() to display the process ID of both parent and child processes.
5. Program Execution using exec()
 - In the child process, call an exec() function (e.g., execl) to execute another program/command.
6. Directory Handling using opendir() and readdir()
 - Open a directory using opendir().
 - Check if the directory is opened successfully.
 - Read directory contents using readdir() in a loop.
 - Display the names of files and directories.
 - Close the directory using closedir().
7. Stop the program.

CODE:

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    pid_t pid = fork();

    if(pid == 0) {
        printf("Child Process ID: %d\n", getpid());
        execl("/bin/ls", "ls", "-l", NULL);
    }
    else {
        printf("Parent Process ID: %d\n", getpid());
        wait(NULL);
    }
    return 0;
}
```

```

}
#include <stdio.h>
#include <dirent.h>

int main() {

    DIR *d;
    struct dirent *dir;

    d = opendir(".");

    if(d) {

        while((dir = readdir(d)) != NULL) {
            printf("%s\n", dir->d_name);
        }

        closedir(d);
    }

    return 0;
}

```

OUTPUT:

```

[hp@localhost ~]$ gcc fork_exec_getpid.c -o fork_exec_getpid
[hp@localhost ~]$ ./fork_exec_getpid
Parent Process: PID = 3125, Child PID = 3126
Child Process: PID = 3126, PPID = 3125
total 36
-rwxrwxr-x. 1 hp  hp  16056 May 23 10:15  fork_exec_getpid
-rw-rw-r--. 1 hp  hp    863 May 23 10:15  fork_exec_getpid.c
-rwxrwxr-x. 1 hp  hp  16104 May 23 10:18  opendir_readdir
-rw-rw-r--. 1 hp  hp    705 May 23 10:18  opendir_readdir.c

[hp@localhost ~]$ gcc opendir_readdir.c -o opendir_readdir
[hp@localhost ~]$ ./opendir_readdir .
Listing of directory: .
.
..
.bash_logout
.bash_profile
.bashrc
.cache
fork_exec_getpid
fork_exec_getpid.c
opendir_readdir
opendir_readdir.c

```

RESULT:

Thus, the system calls `fork()`, `exec()`, `getpid()`, `opendir()`, and `readdir()` were successfully implemented and executed in Linux.